

Information-Theoretic Lower Bounds for Generative/Diffusion Compression with Learned Priors

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Abstract

We derive information-theoretic lower bounds on the compression rate achievable by diffusion-based image codecs (DiffC, PerCo, DiffEIC) that use a pretrained generative model as decoder side information. We extend the classical Wyner-Ziv framework to the case where the side information is a learned neural network prior with bounded KL divergence from the true source distribution. Our main result decomposes the lower bound into three terms: the classical Shannon rate-distortion function $R(D)$, a side-information credit that depends on the prior quality, and a perception premium from the rate-distortion-perception tradeoff. We show that in the ultra-low bitrate regime (< 0.1 bpp), the distortion floor is determined by the prior quality as $D_{\text{floor}} = \sigma^2 \exp(-2\delta)$, where δ is the KL divergence between source and prior.

1 Introduction

Diffusion-based image compression represents a paradigm shift from traditional transform coding. Instead of explicitly encoding pixel data, these methods leverage a pretrained diffusion model as a generative prior and encode only the information needed to guide the generative process toward the target image.

The key question: **what is the minimum bitrate needed when the decoder has a learned generative prior?** Classical information theory provides the Wyner-Ziv rate-distortion function for decoder side information, but assumes the side information is a noiseless observation from a known distribution—not a learned neural network.

2 Problem Setup

Definition 2.1 (Setup). *Let $X \sim p_X$ be the source (e.g., natural images). Let p_{prior} be a learned generative prior (e.g., a diffusion model) available to the decoder. The prior quality is measured by the KL divergence:*

$$\delta = D_{\text{KL}}(p_X \| p_{\text{prior}})$$

The encoder maps X to a bitstream of rate R . The decoder produces $\hat{X}$ using the bitstream and p_{prior} . We seek lower bounds on R subject to $\mathbb{E}[\rho(X, \hat{X})] \leq D$ and $W_2(p_X, p_{\hat{X}}) \leq P$.

3 Main Result

Theorem 3.1 (Diffusion Compression Lower Bound). *The minimum rate for diffusion-based compression with a learned prior of quality δ satisfies:*

$$R \geq R(D) - I_{\text{credit}}(\delta) + \Delta_{\text{perception}}(D, P)$$

where:

$$R(D) = \frac{1}{2} \log_2 \frac{\sigma^2}{D} \quad (\text{classical Shannon bound}) \quad (1)$$

$$I_{\text{credit}}(\delta) = \min\left(\frac{\delta}{\ln 2}, R(D)\right) \quad (\text{side information credit}) \quad (2)$$

$$\Delta_{\text{perception}}(D, P) \geq 0 \quad (\text{RDP perception premium}) \quad (3)$$

Proof Sketch. **Step 1 (Wyner-Ziv extension).** The decoder’s prior acts as side information $S = p_{\text{prior}}$. By the Wyner-Ziv theorem, $R(D|S) \geq I(X; Y) - I(X; S)$.

Step 2 (Prior quality bound). The mutual information between source and prior knowledge is bounded by $I(X; S) \leq \delta / \ln 2$ (in bits), following from the data processing inequality applied to the learned representation.

Step 3 (Perception premium). The perception constraint adds the RDP rate premium (Blau & Michaeli 2019, extended to Gaussian sources in our companion work).

Step 4 (Combination). Combining the three terms gives the lower bound. $\square$

Theorem 3.2 (Ultra-Low Bitrate Regime). *In the ultra-low bitrate regime ($R \ll R(D)$), the minimum distortion is:*

$$D_{\min}(R, \delta) = \sigma^2 \cdot e^{-2\delta} \cdot 2^{-2R}$$

The **distortion floor** (minimum distortion at $R = 0$) is $D_{\text{floor}} = \sigma^2 e^{-2\delta}$.

Corollary 3.3 (Prior Quality Threshold). *A prior with KL divergence δ provides an effective “head start” of $\delta / \ln 2$ bits per sample. Equivalently, the prior reduces the effective source variance from σ^2 to $\sigma^2 e^{-2\delta}$.*

4 Discussion

Our bounds are not tight in general, as the side-information credit is an upper bound and the Gaussian assumption may not hold for natural images. However, the bounds provide:

1. A theoretical benchmark for evaluating diffusion codecs
2. A quantitative relationship between prior quality and achievable bitrate
3. A distortion floor prediction for the ultra-low bitrate regime

References

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